

Cambridge International AS Level
Information Technology 9626

Paper 1 Theory

High-signal questions

110 questions, 10 per topic across the 11 AS Level topics, picked from every Paper 1 question from Feb/March 2022 to May/June 2026. In each topic they cover the points examined most often, favouring longer and more recent questions, and run in syllabus order.

Blank answer lines are left out to save pages; the number in brackets after each part, e.g. [4], is how many marks it is worth.

Each question is numbered (1.1, 1.2, ...) and labelled with the paper it came from. The answers are in "AS Level IT 9626 - Paper 1 high-signal mark schemes.pdf", numbered the same way.

Contents

1	Data processing and information	p. 3
2	Hardware and software	p. 6
3	Monitoring and control	p. 9
4	Algorithms and flowcharts	p. 12
5	eSecurity	p. 19
6	The digital divide	p. 22
7	Expert systems	p. 24
8	Spreadsheets	p. 27
9	Modelling	p. 32
10	Database and file concepts	p. 35
11	Video and audio editing	p. 38

1 Data processing and information

1.1 9626/12 Feb/March 2025 · Question 1

- 1 A college secretary has been given some data from a member of staff.

Chandra, Amruta, YR12538, 59, 75, 68, 04/07/2008

Explain, with examples, how these items of data can become information.

[5]

1.2 9626/12 May/June 2024 · Question 7

- 7 Data can be collected by using a direct or an indirect data source.

Discuss the advantages and disadvantages of using direct data compared to indirect data.

[8]

1.3 9626/12 May/June 2026 · Question 1

- 1 A businesswoman is opening a shop which will sell sports goods. She has decided to research into the spending habits of potential customers.

(a) (i) Describe **one** source of direct data she could use for this purpose.

[1]

(ii) Identify **one** source of indirect data that she could use for this purpose.

[1]

(b) With reference to **both** relevance of data and age of data, explain why the information obtained from the direct data source may produce better quality information.

[4]

The businesswoman has decided to use indirect data for her research into the spending habits of potential customers.

(c) Identify and explain **two** possible reasons, other than relevance of data and age of the data, for her decision to use indirect data for this purpose.

reason 1

reason 2

[4]

1.4 9626/12 Oct/Nov 2022 · Question 8

- 8 Encryption is often used to protect data.

(a) Discuss the need for encryption.

[4]

(b) Discuss the advantages and disadvantages of using symmetric encryption.

[4]

1.5 9626/13 Oct/Nov 2024 · Question 5

- 5 Describe the use of a public key and a private key in asymmetric encryption.

[4]

1.6 9626/12 May/June 2024 · Question 4

- 4 Data can be kept secure by using the Secure Socket Layer (SSL)/Transport Layer Security (TLS) protocols when it is transmitted over the internet.

Describe the use of SSL/TLS to ensure data is kept secure.

[8]

1.7 9626/13 Oct/Nov 2024 · Question 4

- 4 Validation and verification can be used to check the accuracy of data.

- (a) Describe the differences between validation and verification. Your answer must explain the purpose of each.

[3]

- (b) Describe each of the following:

- (i) range check

[2]

- (ii) double data entry

[2]

1.8 9626/13 Oct/Nov 2023 · Question 4

- 4 A bank stores a database of all its customers. The first field in the database is the account number. The account numbers are different for each customer and consist of 10 digits, for example 7382690643.

New accounts are opened in branches of the bank on a regular daily basis.

- (a) Describe **three** errors that a person entering the account numbers of new customers might make. For each one describe an appropriate validation check that would trap the error. A range check or limit check would **not** be suitable. Each validation check must be different.

Error 1

Validation check 1

Error 2

Validation check 2

Error 3

Validation check 3

[6]

When customers open new accounts, the details are gathered together in a transaction file. At the end of the day a branch transmits the transaction file to head office, using the internet.

- (b) The transmission of the transaction file is verified using a hash total.

Describe this process, explaining why a hash total is used rather than a control total.

1.9 9626/11 Oct/Nov 2023 · Question 11

- 11** A company receives orders which are entered into a computer system. The company has decided to use online processing to process these orders so employees enter orders as they are received.

Discuss the advantages and disadvantages of using online processing compared to using batch processing in this scenario.

[8]

1.10 9626/11 May/June 2024 · Question 3

- 3** Describe the computer processing involved in an automatic stock control system.

[7]

2 Hardware and software

2.1 9626/12 Feb/March 2026 · Question 4

- 4 Mainframe computers and supercomputers have differing characteristics.

Using **three** different characteristics, describe differences between mainframe computers and supercomputers.

characteristic 1

difference

characteristic 2

difference

characteristic 3

difference

[6]

2.2 9626/13 Oct/Nov 2023 · Question 3

- 3 Mainframe computers and supercomputers are specialised types of computer with some similar characteristics.

Explain what is meant by each of the following characteristics.

(a) fault tolerance

[3]

(b) heat maintenance

[2]

Mainframe computers were developed from the need for computers to process data in tasks such as carrying out a census.

(c) Explain why mainframe computers are used to process the data from a census.

[3]

2.3 9626/13 May/June 2026 · Question 5

- 5 Supercomputers can be used for weather forecasting.

Discuss, with the example of weather forecasting, the benefits and drawbacks of using a supercomputer.

[6]

2.4 9626/12 May/June 2022 · Question 5

- 5 When a high level language program is written, it has to be translated so the computer can run it. This can be achieved using a compiler or an interpreter.

Discuss the advantages and disadvantages of using a compiler rather than an interpreter.

[8]

2.5 9626/13 May/June 2025 · Question 8

8 Interpreters and device drivers are types of system software.

(a) Describe the functions of device drivers.

[4]

(b) Justify the use of an interpreter.

[4]

2.6 9626/11 Oct/Nov 2024 · Question 1

1 Utility software is needed to maintain a computer system.

Explain why each of the following tasks is needed and how utility software performs each task:

(a) disk formatting

[3]

(b) data compression

[3]

(c) disk defragmentation

2.7 9626/13 Oct/Nov 2022 · Question 3

3 Back-up software and anti-virus software are both types of utilities that are often provided with operating systems.

(a) Without detailing the different methods of detection, describe the features of anti-virus software.

[3]

(b) Without referring to the different types of back-up, describe the features of back-up software.

[3]

2.8 9626/12 Feb/March 2026 · Question 9

9 Wei has recently started an accountancy business. She is considering using either custom-written or off-the-shelf software in the business.

Discuss the suitability of each type of software.

[8]

2.9 9626/13 May/June 2023 · Question 9

9 Discuss the advantages and disadvantages of using a gesture-based user interface rather than a graphical user interface to control devices.

[8]

2.10 9626/11 May/June 2026 · Question 9

9 An international airport has installed a computer system to record customer satisfaction. The system uses a graphical user interface (GUI) involving the use of graphical icons.

(a) Justify the use of the GUI for this computer system.

[6]

- (b) Identify and explain **two** reasons why a dialogue interface may **not** be suitable for this system.

[4]

3 Monitoring and control

3.1 9626/12 May/June 2022 · Question 6

- 6 Sensors are used in monitoring systems to measure physical variables. It is important that sensors are calibrated before use.
- (a) Describe, in detail, these sensors and give an appropriate use for each. Each use must be different.
- Humidity
- Sound
- [4]
- (b) Explain the importance of calibration.
- [2]
- (c) Describe the differences between one point calibration and two point calibration.
- [5]

3.2 9626/11 and 9626/13 Oct/Nov 2025 · Question 5

- 5 (a) Explain the reasons why sensors must be calibrated before they are used.
- [2]
- (b) Compare one-point calibration with multipoint calibration.
- [5]

3.3 9626/11 May/June 2025 · Question 7

- 7 Sensors can be used to collect data about physical variables over a period of time.
- (a) Describe the use of different sensors to collect data about the weather.
- [6]
- (b) Describe how the data from the sensors can be processed.
- [6]

3.4 9626/13 May/June 2024 · Question 6

- 6 The environmental conditions around a chemical factory need to be monitored to make sure the environment is safe from pollution.
- (a) Describe how data logging could be used to help the factory owner monitor the environmental conditions around the factory.
- [6]
- (b) Discuss the advantages and disadvantages of monitoring the environmental conditions around the factory using a computer system rather than using manual methods.
- [8]

3.5 9626/11 Oct/Nov 2023 · Question 6

- 6 Most control technology systems use actuators and sensors.
- (a) Describe the purpose of an actuator. [2]
- (b) Describe how an electromagnetic field sensor works when used in a car park control technology system. [2]
- (c) Describe how an ultrasonic sensor works when a driver is parking a car. [2]

3.6 9626/12 Feb/March 2024 · Question 5

- 5 An apartment has a controlled central heating system. This system includes a microprocessor.
- Explain how the temperature in the apartment is controlled by the system. [6]

3.7 9626/11 Oct/Nov 2024 · Question 8

- 8 (a) Describe the purpose of **three** sensors that are needed in a microprocessor-controlled burglar alarm system. [3]
- (b) Explain how a microprocessor-controlled burglar alarm system works. [3]

3.8 9626/13 May/June 2024 · Question 9

- 9 A car park barrier opens automatically when a vehicle approaches and closes when the vehicle is clear of the barrier.
- Describe how a microprocessor could be used to control the operation of the barrier. [6]

3.9 9626/12 May/June 2023 · Question 6

- 6 Sensors are used in monitoring and control to measure physical variables.
- (a) Describe these sensors. For each sensor, explain how it could be used in a control system. Each system must be different.
- Infrared
- Ultrasonic [4]
- (b) Control technologies are replacing human-controlled car park barriers and timer-controlled traffic lights.
- Discuss the advantages and disadvantages of the use of control technologies in these scenarios.

3.10 9626/11 May/June 2025 · Question 4

- 4** Describe the drawbacks of using microprocessors to control devices and appliances in a smart home.

4 Algorithms and flowcharts

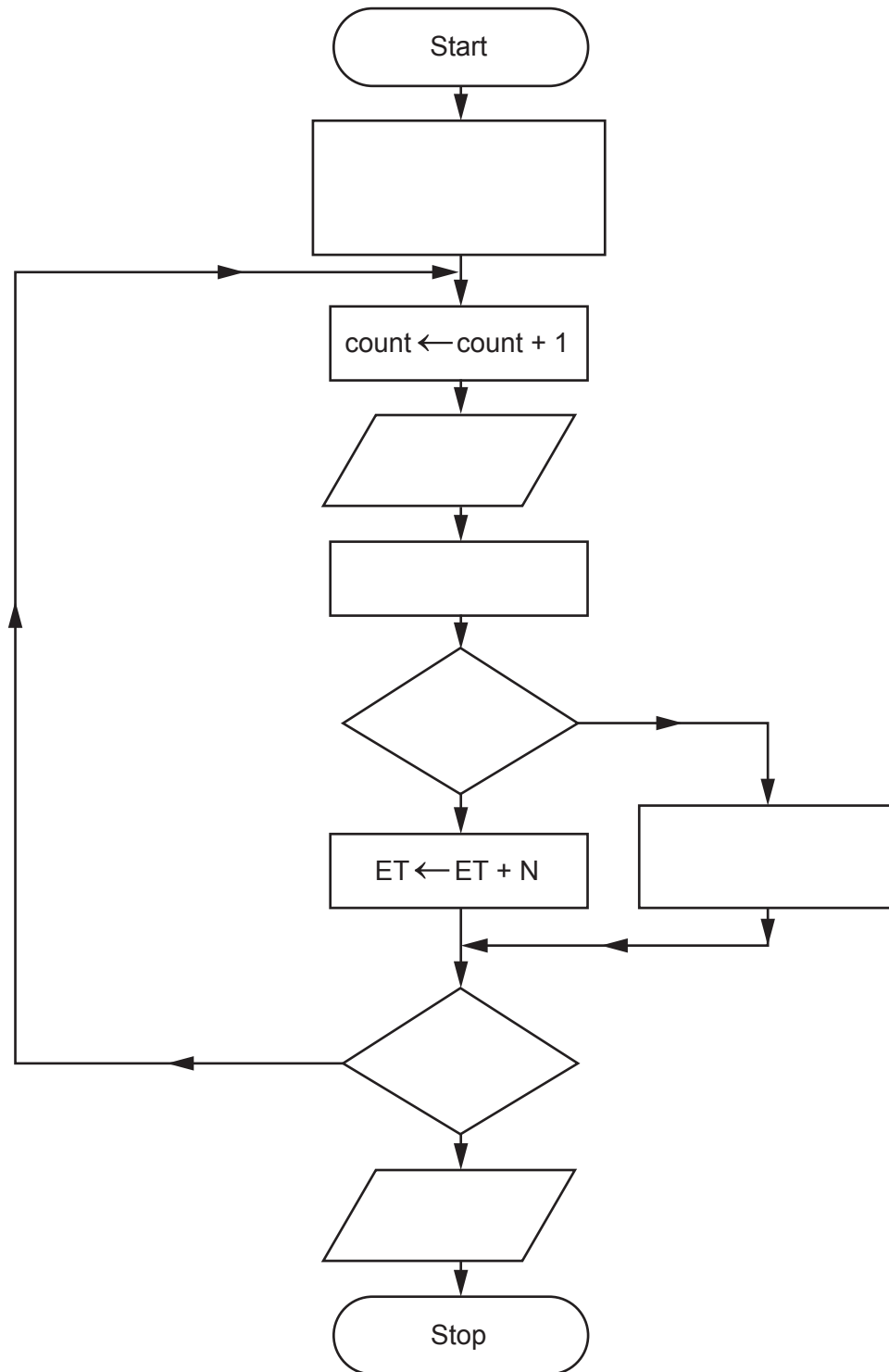
4.1 9626/12 Oct/Nov 2024 · Question 4

- 4 Draw a program flowchart to represent this algorithm:
- Input an integer value.
 - Repeat this until you have entered 20 values.
 - At the end, output the largest value and the average of the values.

[7]

4.2 9626/11 May/June 2023 · Question 8

- 8 Complete this flowchart which will allow the input of ten numbers.
- A number is even if the number divided by 2 is an integer.
 - If a number (N) is odd, it will be added to the odd total (OT).
 - If a number (N) is even, it will be added to the even total (ET).
 - Both these totals will be output after the tenth number has been processed.



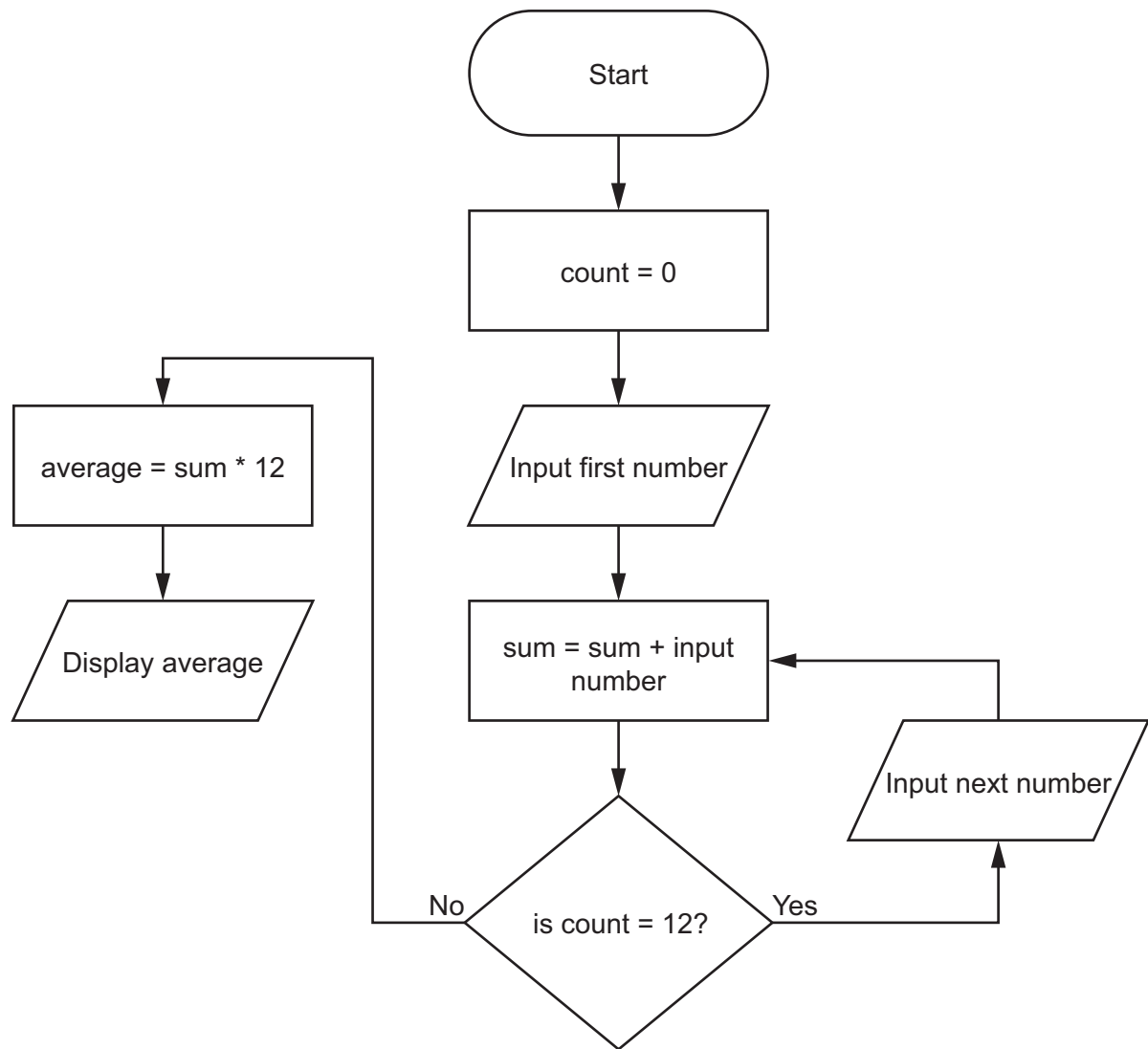
[8]

4.3 9626/13 May/June 2025 · Question 13

13 The average (mean) of a set of 12 numbers is calculated by:

- adding the numbers together to get a total
- dividing the total by 12.

A student has attempted to draw a flowchart of an algorithm which finds the average of 12 numbers. They have made some errors.



Redraw the flowchart, with the errors corrected.

[5]

4.4 9626/12 Feb/March 2022 · Question 7

- 7 The average (mean) of a set of numbers can be found by adding the numbers together and dividing the resulting total by how many numbers there are in the set.

Write an algorithm in pseudocode to enter and find the average of a set of numbers using a REPEAT...UNTIL loop. Your algorithm must work for different sets of numbers.

[8]

4.5 9626/13 May/June 2025 · Question 7

- 7 A teacher is writing a program to process test marks for a class.

For marks above 70, the program will output "You have achieved a Distinction".
 For marks from 55 to 70, the program will output "You have achieved a Merit".
 For marks from 40 to 54, the program will output "You have achieved a Pass".
 For marks below 40, the program will output "You have not reached the Pass mark".

Write a pseudocode algorithm using REPEAT...UNTIL which allows the teacher to input 20 test marks and outputs the correct message for each inputted mark.

[8]

4.6 9626/11 Oct/Nov 2023 · Question 7

7 A company employs workers as:

- labourers who are paid \$18 per hour
- office workers who are paid \$22 per hour
- managers who are paid \$30 per hour.

A computer programmer has written part of an algorithm in pseudocode. The completed algorithm will calculate and output the wage for each worker.

The inputs will be:

- the number of workers
- the job code for each worker (L for labourer, O for office worker or M for manager)
- the number of hours each worker has worked.

If L, O or M is not entered for the job code then the message 'Invalid job code' will be output. The algorithm will terminate when all the workers' details have been entered. It will use the CASE...ENDCASE construct to determine the rate of pay.

Complete the algorithm by filling in the missing lines.

INPUT number

REPEAT

```
INPUT jobcode, hours
```

```
'L' : rate ← 18
```

```
count ← count + 1
```

[8]

4.7 9626/12 May/June 2026 · Question 2

2 A partially completed algorithm is shown. Line numbers are given for reference. Some lines are incomplete.

The user will input five numbers, one number at a time.

After five numbers have been entered, the largest number and the smallest number will be displayed as part of a suitable message.

The algorithm uses a WHILE loop.

```

10  Count  ←    0
20  Number ←    0
30  Largest_Number ← 0
40  Smallest_Number ← .....
50  INPUT Number
60  Largest_Number ← Number
70  Smallest_Number ← Number
80  WHILE Count < .....
90      INPUT .....
100     IF Number ..... Largest_Number
110         .....
120             Largest_Number ← Number
130         .....
140             IF Number ..... Smallest_Number
150                 THEN Smallest_Number ← Number
160             ENDIF
170         .....
180     Count  ←    .....
190 ENDWHILE
200 PRINT "The largest number is "Largest_Number"
      and the smallest number is "Smallest_Number"."

```

- (a) Use the table to show how each line may be completed. Line 40 has been done for you as an example of how the question should be answered.

line	current code	completed code
40	Smallest_Number ←	Smallest_Number ← 0
80	WHILE Count <	
90	INPUT	
100	IF Number Largest_Number	
110		
130		
140	IF Number Smallest_Number	
170		
180	Count ←	

[8]

- (b) Explain why Largest_Number (line 60) and Smallest_Number (line 70) are both set to the same value collected in line 50.

[4]

4.8 9626/12 Feb/March 2026 · Question 7

7 A student has created an algorithm to calculate the average of a set of numbers.

- The user will be asked how many numbers they wish to enter
- The user will respond with a number
- The user will then enter that many numbers
- The algorithm will total the numbers entered and divide the total by how many numbers the user said they will enter.

The algorithm is shown. The line numbers have been included to help you.

```
10 total ← 10
20 PRINT "Input number of repeats"
30 INPUT n
40 count → 1
50 REPEAT
60     count ← + 1
70     PRINT "Input number ", count
80     INPUT number
90     total + number ← total
100 UNTIL count = number
110 average ← total / n
120 PRINT average
```

The student has made some mistakes in the algorithm.

(a) State **three** mistakes in this algorithm and suggest how each can be corrected.

- mistake 1
- mistake 2
- mistake 3

[6]

This algorithm uses a REPEAT...UNTIL loop. The student wishes to change the algorithm so that it uses a WHILE...ENDWHILE loop.

(b) Contrast a REPEAT...UNTIL loop with a WHILE...ENDWHILE loop.

[2]

4.9 9626/12 Oct/Nov 2022 · Question 9

9 A programmer is writing an algorithm in pseudocode to represent a company's payroll system. The system calculates a worker's wages before tax by multiplying the number of hours worked by the rate of pay per hour. A procedure within the main algorithm representing this could be:

```
PROCEDURE BeforeTax(Hours, Rate)
    WagesBeforeTax ← Hours * Rate
ENDPROCEDURE
```

There are two stages involved in calculating the wages after tax:

- the amount of tax paid by the worker is calculated by multiplying the WagesBeforeTax by the rate of tax (35%);
- the amount of tax paid is then subtracted from the WagesBeforeTax.

(a) Write a procedure for calculating the wages after tax, assuming the value of WagesBeforeTax is passed to it.

[4]

- (b) The programmer's algorithm will use the `BeforeTax()` procedure and the procedure created in part (a) to calculate and output the wage after tax for each worker.

Complete the algorithm. The statements have been numbered to help you.

```
1  count ← 0
2  INPUT NumberOfWorkers
3  REPEAT
4  .....
5  .....
6  .....
7  .....
8  .....
9  UNTIL ..... [6]
```

4.10 9626/11 May/June 2022 · Question 6

- 6 Many houses in cooler countries have central heating systems. In a central heating system, microprocessors are used to control the pump which sends water from the boiler to the individual heaters.

Complete the pseudocode algorithm to show the processing which takes place in this microprocessor-controlled central heating system. The algorithm must prevent attempting to switch the pump on when it is already on and attempting to switch the pump off when it is already off. You may assume `temperature` is the variable representing the actual temperature of the house and `preset` represents the required temperature of the house.

```
REPEAT
INPUT temperature
  IF temperature < preset
    THEN
```

[6]

5 eSecurity

5.1 9626/12 May/June 2023 · Question 4

- 4 Some individuals post personal data on social media or in emails.

Discuss why personal data needs to be kept confidential.

[8]

5.2 9626/12 Feb/March 2026 · Question 1

- 1 Irene would like to post a photograph of her family on a social media website.

- (a) Describe **two** ways in which Irene can ensure that the personal data in the photograph can be protected after it has been posted online.

way 1

way 2

[2]

Irene has installed a firewall on her home computer network.

- (b) (i) Describe how a firewall works to prevent unauthorised access to a computer network.

[4]

- (ii) Describe **one** drawback of using a firewall on a computer network.

[1]

5.3 9626/12 Oct/Nov 2022 · Question 3

- 3 Phishing, smishing and vishing are methods of fraudulently obtaining personal data.

- (a) Describe **two** differences between phishing and vishing.

[4]

- (b) Describe how smishing is used to obtain personal data.

[3]

5.4 9626/12 Feb/March 2022 · Question 8

- 8 Phishing is an attempt by fraudsters to gather personal information using email. Receivers of emails should use several different methods to prevent phishing attacks.

Evaluate the different methods which can be used to prevent phishing.

[8]

5.5 9626/12 May/June 2022 · Question 8

- 8 Pharming is a method used by hackers to obtain information from bank customers.

- (a) Explain what is meant by pharming.

[4]

- (b) Describe **three** methods a customer could use to minimise the risk of becoming a victim of pharming.

[3]

5.6 9626/12 May/June 2024 · Question 6

- 6 Describe these types of malware.

(a) Malicious bots

[4]

(b) Rootkit

[4]

(c) Virus

[4]

5.7 9626/12 Feb/March 2023 · Question 9

- 9 Describe what is meant by the malware terms.

(a) worm

[4]

(b) Trojan

[4]

5.8 9626/12 Feb/March 2022 · Question 9

- 9 Hackers sometimes commit fraud after causing spyware to be installed on other users' computers.

(a) Describe what is meant by the term 'spyware'.

[3]

(b) Explain how a hacker uses the information gained from spyware to commit fraud.

[4]

5.9 9626/11 May/June 2026 · Question 7

- 7 A company has been hit by a ransomware attack.

(a) Describe the possible effects of this ransomware attack on the company.

[2]

(b) Evaluate the use of physical methods to prevent a second ransomware attack on the company.

[6]

5.10 9626/12 May/June 2025 · Question 6

- 6** Computer systems can be affected by malware. A network manager has already installed malware prevention software.

Describe additional methods the network manager can use to help prevent malware from entering the computer systems.

[6]

6 The digital divide

6.1 9626/13 Oct/Nov 2022 · Question 10

- 10 Describe, listing the groups affected, what is meant by the digital divide.

[6]

Please turn over for Question 11.

6.2 9626/13 Oct/Nov 2024 · Question 8

- 8 Describe how access to information technology has created the digital divide.

[8]

6.3 9626/13 May/June 2026 · Question 7

- 7 A young adult and his grandfather use modern technology in different ways.

Discuss possible reasons why the young adult may use modern technology to a greater extent than his grandfather.

[8]

6.4 9626/12 Oct/Nov 2024 · Question 5

- 5 Explain the reasons why there is a digital divide between people with high levels of education and people with lower levels of education.

[6]

6.5 9626/12 May/June 2023 · Question 7

- 7 Explain why there can be a digital divide between different socioeconomic groups.

[5]

6.6 9626/12 Oct/Nov 2023 · Question 6

- 6 There can be a digital divide between people who live in rural areas and people who live in urban areas. The divide can be caused by the lack of fast broadband in rural areas, resulting in limited access to the web.

(a) Explain the effects on people who live in rural areas of having limited access to the web.

[4]

(b) Describe ways to reduce this digital divide.

[3]

6.7 9626/12 Feb/March 2024 · Question 12

- 12 A digital divide exists between the more technologically aware and the less technologically aware nations.

Analyse this type of digital divide.

[6]

6.8 9626/12 Oct/Nov 2025 · Question 13

- 13** Describe the effects of the digital divide on health services.

[6]

6.9 9626/12 May/June 2024 · Question 3

- 3** A government is concerned about the digital divide amongst its citizens.

Describe how the government could reduce the effects of the digital divide and improve access to its services.

[6]

6.10 9626/11 and 9626/13 Oct/Nov 2025 · Question 14

- 14** Identify **two** groups of people who can be negatively affected by the digital divide. For each group, suggest a way the digital divide can be reduced.

[4]

7 Expert systems

7.1 9626/11 Oct/Nov 2024 · Question 10

- 10** A garage uses an expert system to diagnose faults in car engines.

Describe the components of this expert system.

[8]

7.2 9626/12 Oct/Nov 2025 · Question 3

- 3** An expert system contains an inference engine and a knowledge base.

(a) Describe how the inference engine performs its role.

[4]

(b) Describe how a knowledge base is created.

[4]

7.3 9626/13 May/June 2024 · Question 3

- 3** In an expert system, the inference engine uses reasoning that consists of backward or forward chaining.

Explain the differences between backward and forward chaining.

[6]

7.4 9626/13 May/June 2023 · Question 10

- 10** An expert system is being trialled for use as part of a study on animal identification. It has been set the following task.

The goal is to decide the colour of a bird named Hortense.

Two facts are provided:

- she sings
- she is 15 cm tall.

The rule base contains the following three rules:

- IF X is an ostrich – THEN X has a long neck
- IF X sings and X is 15 cm tall – THEN X is a canary
- IF X is a canary – THEN X is yellow

The inference engine has concluded that Hortense is yellow. It has taken the facts as data and used the rules base to decide:

- 1 Hortense is a canary
- 2 Hortense is yellow

(a) Identify which type of chaining the inference engine used.

[1]

(b) Explain the reasoning the inference engine used.

[4]

7.5 9626/12 Feb/March 2026 · Question 6

- 6 An expert system may be data driven.

Describe, with the use of an example, what is meant by the term data driven.

[5]

7.6 9626/13 Oct/Nov 2024 · Question 7

- 7 Describe the term goal driven as used in expert systems.

[4]

7.7 9626/12 May/June 2023 · Question 9

- 9 A doctor uses an expert system to help the doctor diagnose the illnesses of her patients.

Explain how an expert system is used to help the doctor produce the diagnosis of an illness. You may assume only forward chaining is involved.

[8]

7.8 9626/11 May/June 2026 · Question 2

- 2 A company employs drivers to deliver parcels in a city. The company uses an expert system to plan the routes each driver will follow each day.

Describe how an expert system can be used to provide route scheduling in this scenario.

[8]

7.9 9626/11 May/June 2024 · Question 5

- 5 A bank uses an expert system to help customers with financial planning.

(a) Describe the use of **four** components of an expert system. Your answers **must** relate to financial planning.

[4]

(b) Describe the benefits, to the bank, of using an expert system.

[6]

(c) State **three** drawbacks, to the bank, of using an expert system.

[3]

7.10 9626/12 May/June 2026 · Question 7

- 7 A mechanic uses an expert system.

This system has a user interface and an explanation system.

(a) Identify and describe **two** other components of an expert system.

component 1

component 2

[4]

The expert system is used to diagnose faults in a car's engine.

(b) Evaluate the use of this system.

[8]

8 Spreadsheets

8.1 9626/12 May/June 2026 · Question 3

- 3 A student has created a spreadsheet to store data.

(a) Describe **two** features of spreadsheet software that can be used to make the process of entering data into the spreadsheet easier for the student.
feature 1
feature 2

[2]
- The student needs to analyse the information she has entered.

(b) Describe **four** features of spreadsheet software that can be used to analyse the information that has been entered into the spreadsheet.

[4]

8.2 9626/13 May/June 2024 · Question 4

- 4 Describe the terms absolute and relative cell replication, giving an example of each.
Absolute
Relative

8.3 9626/12 May/June 2025 · Question 2

- 2 The spreadsheet shows the prices of various items in a clothes shop. The discounted price for a jumper is calculated by using the formula in cell F12.

F12		✕ ✓ fx		=E12-(VLOOKUP(D12,\$D\$5:\$E\$8,2,FALSE)*E12)				
	A	B	C	D	E	F	G	H
1								
2								
3				Discount code				
4								
5				A	25%			
6				B	20%			
7				C	10%			
8				D	5%			
9								
10	Item				Price	Discounted price		
11								
12	Jumper			B	\$36.00	\$28.80		
13	Skirt			A	\$25.00	\$18.75		
14	Blouse			D	\$30.00	\$28.50		
15	Trousers			A	\$40.00	\$30.00		
16	Jeans			C	\$33.00	\$29.70		
17	Dress			B	\$50.00	\$40.00		
18	Trainers			A	\$55.00	\$41.25		
19	Shoes			D	\$60.00	\$57.00		
20								

- (a) The formula in cell F12 is =E12-(VLOOKUP(D12,\$D\$5:\$E\$8,2,FALSE)*E12)

Describe how this formula works.

[5]

The formula in cell F12 is replicated to cells F13 to F19.

(b) In the formula, absolute cell referencing and relative cell referencing have **both** been used.

Explain why both types of cell referencing are needed.

[4]

8.4 9626/13 Oct/Nov 2024 · Question 6

6 In a spreadsheet, a user has entered the following formula into a cell:

=IF(NOT(OR(B6<5,B6>8)),"4th Quarter","")

Describe how the spreadsheet will calculate the result to be displayed.

[3]

8.5 9626/11 May/June 2026 · Question 3

3 A teacher is creating a spreadsheet.

(a) Describe **two** ways in which the spreadsheet can validate data at the input stage.

[2]

The spreadsheet will include both absolute cell referencing and relative cell referencing.

(b) Define the term absolute cell referencing.

[1]

(c) Identify and explain **one** benefit of using relative cell referencing when creating a spreadsheet.

[2]

(d) The teacher has used the following formula in a cell:

=IF(C2="Dog",(IF(D2="Male","Block A","Block B")),(IF(D2="Male","Block C","Block D")))

Describe how the spreadsheet will produce the result to be displayed.

[6]

8.6 9626/11 May/June 2025 · Question 12

12 A spreadsheet is to be used to store students' examination results.

Each examination is marked out of 60, with the lowest mark being 0.

Only whole marks are allowed.

Each mark is validated using this validation rule ≥ 0 and ≤ 60

The developer is trying to test the validation rule.

Complete the table for the **three** types of test data. For each mark the developer enters, suggest how the validation rule deals with the mark.

Type of test data	Example of mark entered	How the validation rule deals with the mark entered
Normal		
Abnormal		
Extreme		

[6]

8.7 9626/13 May/June 2025 · Question 5

- 5 A student created a spreadsheet that requires users to enter numerical data. This data must be within a set range.

Identify and justify **three** types of test data that can be used to test this numerical data.

[6]

8.8 9626/13 May/June 2026 · Question 8

- 8 A teacher has created a spreadsheet to store marks and grades achieved by students in an examination. Part of this spreadsheet is shown.

	A	B	C	D	E	F	G	H
1	Grade boundaries				Student grades			
2	Grade	Minimum Mark	Maximum Mark		Student name	Score	Grade	
3	A	44	50		Kyle			
4	B	40	43		Germain			
5	C	36	39		Jose			
6	U	0	35		Sandy			
7								

- (a) Describe the formatting that has been applied to the contents of cell B2.

[2]

The teacher needs to test the spreadsheet. When testing a spreadsheet, it is important to use appropriate test data.

(b) Describe **one** defining characteristic of each of:

- normal test data
- extreme test data
- abnormal test data.

[3]

(c) Complete the table below to identify **three** items of test data that can be used to test the mark range for a grade A.

test type	test data
normal	
extreme	
abnormal	

[3]

	A	B	C	D	E	F	G	H
1	Grade boundaries				Student grades			
	Grade	Minimum Mark	Maximum Mark		Student name	Score	Grade	
2								
3	A	44	50		Kyle			
4	B	40	43		Germain			
5	C	36	39		Jose			
6	U	0	35		Sandy			
7								

The teacher will use the following formula in cell G3 to show the grade achieved by Kyle.

=IF(F3>C3,"Check Score",(IF(F3>B3,A3,(IF(F3>B4,A4,(IF(F3>B5,A5,A6)))))))

(d) This formula has been tested and found not to work as intended.

Explain why this formula does not work as intended.

[5]

8.9 9626/12 Feb/March 2025 · Question 7

- 7 Test scores can be used to track a student's progress over time. Describe **two** ways this can be done using a spreadsheet.

Way 1
Way 2

[4]

8.10 9626/11 May/June 2025 · Question 2

- 2 The manager of a gymnasium wishes to keep records of all the members' details including performance results. She is unsure whether to use a spreadsheet or a database.

Discuss the benefits and drawbacks of using a spreadsheet to undertake this task.

9 Modelling

9.1 9626/12 Oct/Nov 2024 · Question 6

- 6 Describe the reasons why computer models are used.

[6]

9.2 9626/12 Feb/March 2026 · Question 8

- 8 Identify and describe **three** characteristics of modelling software.

characteristic 1

characteristic 2

characteristic 3

[6]

9.3 9626/12 Oct/Nov 2022 · Question 4

- 4 Here is a spreadsheet created by the organisers of a concert. They have assumed that 400 people will attend the concert. They have also assumed that each member of the audience will purchase 2 snacks and 2 drinks during the concert. If these assumptions are correct, their profit will be \$710.

	A	B	C	D
1		Estimated audience size		400
2				
3		Price	Quantity per concert goer	Income
4	Tickets	\$40	1	\$16,000
5	Drinks	\$2	2	\$1,600
6	Snacks	\$3	2	\$2,400
7	Souvenirs	\$20	1	\$8,000
8	Programmes	\$15	1	\$6,000
9			Total income	\$34,000
10				
11				Paid out
12	Electricity			\$650
13	Performers			\$22,000
14	Drink supplies			\$860
15	Snack supplies			\$880
16	Venue hire			\$5,000
17	Souvenir supplies			\$2,400
18	Programme supplies			\$1,500
19			Total outgoings	\$33,290
20				
21			Profit/Loss	\$710
22				

A number of factors could affect the amount of profit.

- The audience size might be higher than the organisers estimated.
- Members of the audience might not buy 2 snacks and 2 drinks.
- The cost of souvenir supplies might change.

- (a) Explain how the organisers could use what-if analysis to see the effects of these three factors on their profit. You should refer to specific cells from the spreadsheet in your answer.

[3]

- (b) The organisers wish to achieve a larger profit.

By explaining what each step requires them to do, describe how the organisers would use goal seek to find the size of the audience needed to make a profit of \$5000. You should refer to specific cells from the spreadsheet in your answer.

Include in your answer a reason why the result might not be satisfactory.

[4]

9.4 9626/11 May/June 2026 · Question 4

- 4 Computer models make use of what-if analysis.

- (a) Describe, with the use of relevant examples, how what-if analysis can be used as part of a model of population growth.

[4]

- (b) Evaluate the use of a computer simulation as a suitable method to plan a response to a natural disaster.

[8]

9.5 9626/13 May/June 2025 · Question 11

- 11 Evaluate the use of spreadsheets for financial modelling.

[8]

9.6 9626/12 Oct/Nov 2025 · Question 4

- 4 Identify and explain **two** drawbacks of using a spreadsheet for computer modelling.

[4]

9.7 9626/13 Oct/Nov 2023 · Question 11

- 11 Weather forecasting makes use of computer models.

Analyse the effectiveness of computer models for producing weather forecasts.

[8]

9.8 9626/11 Oct/Nov 2023 · Question 9

- 9 Computer models are often used to run simulations when designing a nuclear power plant.

- (a) Describe the benefits of using simulations in this scenario.

[4]

- (b) Describe the drawbacks of using simulations in this scenario.

[4]

9.9 9626/11 May/June 2024 · Question 6

- 6 An airline company uses a flight simulator to train its pilots.

Discuss the advantages and disadvantages of using a flight simulator in this scenario.

9.10 9626/12 Feb/March 2025 · Question 11

- 11** A driving school is using computer simulations to improve its lessons.

Discuss the benefits and drawbacks of using simulators for this purpose.

10 Database and file concepts

10.1 9626/12 May/June 2024 · Question 1

- 1 A building company has a relational database. The database has two tables: CUSTOMERS and QUOTATIONS.
- (a) The QUOTATIONS table has a foreign key.
- Describe what is meant by the term foreign key.
- [2]
- (b) (i) Define the term referential integrity.
- [1]
- (ii) Explain why referential integrity is important.
- [3]
- (c) Describe what is meant by the term relational database.
- [5]

10.2 9626/11 Oct/Nov 2024 · Question 4

- 4 Relationships are used in relational databases.
- (a) Describe the following database relationships:
- (i) one-to-one
- [1]
- (ii) one-to-many
- [1]
- (b) Describe how a many-to-many relationship could be created in a database.
- [3]

10.3 9626/13 May/June 2026 · Question 9

- 9 A student is planning a relational database.
- (a) Define the term relational database.
- [1]
- The relational database could include both primary keys and compound keys.
- (b) (i) Describe the characteristics of a primary key in a relational database.
- [2]
- (ii) Describe the role of a primary key in a relational database.
- [1]
- (c) Describe how a compound key differs from a primary key.
- [1]

The database the student is planning needs to be normalised.

(d) Describe the conditions that must be met for the database to be in first normal form (1NF). [4]

(e) Describe **four** benefits of normalising the relational database. [4]

10.4 9626/12 May/June 2023 · Question 8

8 Peter wants to create a database containing the details of all the books he owns and the shops he bought them from. Mary suggests that he should create a relational database. Peter will need to use normalisation.

(a) Describe what is meant by normalisation. [4]

(b) Describe the benefits of creating a relational database rather than a flat file database. [6]

10.5 9626/12 Feb/March 2024 · Question 2

2 A data dictionary is required when creating a relational database.

Identify and describe **four** components that should be included in a data dictionary. [8]

10.6 9626/11 May/June 2024 · Question 9

9 A company in Nairobi, Kenya, uses a database to store all its customer details.

Evaluate the use of static and dynamic parameters in a query to search the database for customers in Nairobi, Kenya. [8]

10.7 9626/11 May/June 2025 · Question 10

10 Describe the following types of database query:

(a) summary [2]

(b) complex [2]

(c) nested [2]

10.8 9626/12 Oct/Nov 2023 · Question 8

8 Students often transfer files between different word processing and database packages.

Justify the need for generic file formats for the transfer of files between different packages. [8]

10.9 9626/11 May/June 2024 · Question 4

- 4 Describe the term indexed sequential access.

[8]

10.10 9626/11 Oct/Nov 2024 · Question 3

- 3 Describe how a company can use management information system (MIS) software to help run its business.

[6]

11 Video and audio editing

11.1 9626/12 Oct/Nov 2023 · Question 10

10 Explain why the following techniques could be needed when making videos.

- (a) trimming a video clip [2]
- (b) adding subtitles to a video [2]
- (c) adding sound to a video [2]

11.2 9626/13 May/June 2024 · Question 8

8 Explain why the following techniques could be used when editing a video.

- (a) Creating credits [2]
- (b) Extracting a still image [2]
- (c) Exporting a video clip in a different file format [2]

11.3 9626/11 May/June 2026 · Question 5

5 A video has been created with an aspect ratio of 16:9.

- (a) Describe what is meant by an aspect ratio of 16:9. [1]
- (b) Explain **two** reasons why an aspect ratio of 16:9 rather than an aspect ratio of 4:3 has been chosen. [2]

11.4 9626/12 Oct/Nov 2025 · Question 8

8 There are two methods of file compression, one is Lossy the other is Lossless.

- (a) Describe what is meant by Lossy compression. [1]
- (b) Describe what is meant by Lossless compression. [1]
- (c) Compare the impact on quality of video of using each of these methods of compression. [4]

11.5 9626/12 May/June 2024 · Question 8

8 Sound editing software has a number of features that can be used to edit audio files.

(a) State **two** reasons why an audio clip may need to be trimmed.

Reason 1

Reason 2

[2]

(b) State **one** reason why separate audio clips may need to be spliced together.

[1]

(c) Describe the term equalisation when applied to audio clips.

[3]

11.6 9626/12 Feb/March 2025 · Question 10

10 Audio editing software has many features. These can be used when editing an audio clip.

Describe the following terms when applied to audio clips:

(a) normalisation

[2]

(b) noise reduction.

[2]

11.7 9626/11 and 9626/13 Oct/Nov 2025 · Question 8

8 Sound files can be edited.

Describe the following features. For each feature, give a reason why it is used.

(a) overdubbing

[2]

(b) equalisation

[2]

(c) reverberation

[2]

11.8 9626/13 Oct/Nov 2023 · Question 7

7 Sound editing software has a number of features.

Describe what is meant by:

(a) fade in and fade out

[2]

(b) lossless compression

[2]

(c) lossy compression.

[2]

11.9 9626/12 Feb/March 2024 · Question 10

10 Sound editing software can be used to alter the sampling rate in audio files.

(a) Explain what is meant by the term sampling rate.

[2]

(b) Explain how the sampling rate can affect the audio files.

[2]

11.10 9626/11 May/June 2023 · Question 10

10 The quality of sound produced from the data stored on an optical disk depends on the sampling resolution.

Describe what is meant by sampling resolution.

[5]